K-102
BPA DUST COLLECTOR
CAPACITY: 25 M3/HR
DP: 250 MMWCG/FV
DT: 60 °C
MOC: SS304 (WETTED PARTS)
(BY K-101 VENDOR)

K-101A/B
BPA HOPPER
CAPACITY: 2 M3
DIM: 1500 MM ID X 1100 MM T-T
DP: +250/-200 MMWCG
DT: 100 °C
MOC: SS304
INS: NONE

K-103
BPA HOIST-I

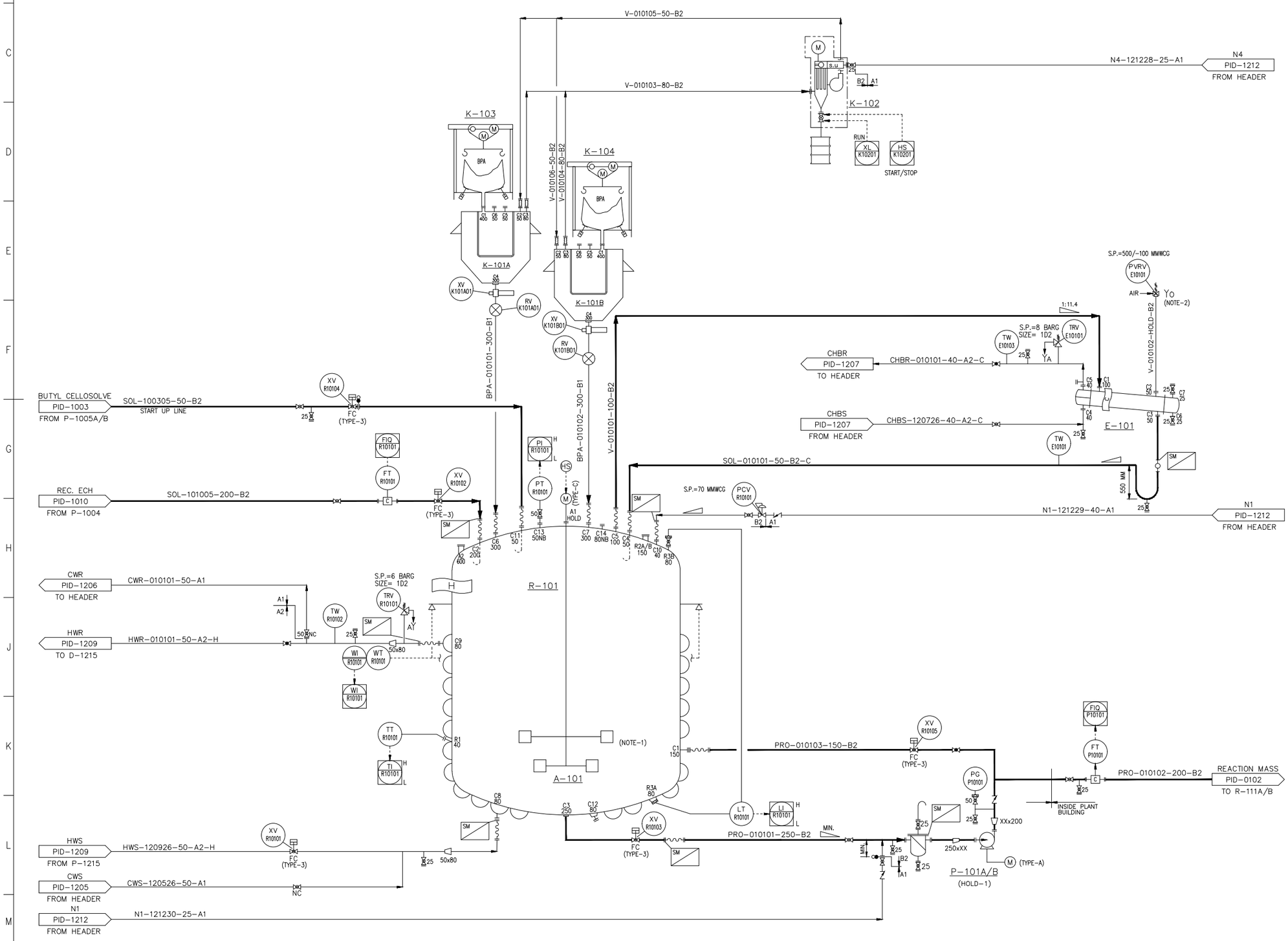
K-104
BPA HOIST-II

A-101
PREMIX REACTOR AGITATOR
TYPE: TOP ENTRY
MOC: SS304L
MOTOR: HOLD KW

R-101
PREMIX REACTOR
CAPACITY: 46 M3
DIM: 3700 MM ø x 3700 MM T-T
DP(V//C): 1/-500 MMWCG//6 BAR G
DT(V//C): 100/5//120/5 °C
MOC(V/C): SS304L/SS304L
INS: HOT

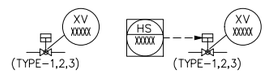
P-101 A/B
PREMIX REACTOR PUMP
CAPACITY: 132 M3/H @ TDH 23.3 M
POWER REQ'T: 9 KW (HYD.)
MOC(CASING/IMPELLER): SS304L/SS304L
INS: NONE

E-101
PREMIX REACTOR CONDENSER
TYPE: SHELL & TUBE(BEM)
DUTY: 15805 KCAL/HR
DP(S/T): 6.2/8 BAR G
DT(S//T): 100/0//60/-10 °C
MOC(S/T): SS304/SS304
INS(S/T): COLD/COLD



GENERAL NOTES:

1. ALL VAPOR LINES, SLURRY LINES AND GRAVITY LINES SHOULD HAVE MINIMUM BENDS AND NO LOOPS.
2. PROVIDE PROPER EARTHING AND BONDING FOR THE HYDROCARBON LINES.
3. DRAIN TO BE PROVIDED AT THE LOWEST POINT AND VENT TO BE PROVIDED AT THE HIGHEST POINT.
4. REFER LEGEND SHEET FOR ABBREVIATIONS, SYMBOL, ON-OFF VALVE DETAILS, MOTOR/VFD AND STEAM TRAP DETAILS.
   DWG NO. EPAM01-PR-PID-1000,
   EPAM01-PR-PID-1001,
   EPAM01-PR-PID-1002.
5. FOR LINES WITH SLOPE, SLOPE TO BE PROVIDED IN SUCH A WAY THAT THE LINE SHOULD BE DRAINED ON EITHER SIDE.
6. PROPOSED SPECIAL PIPING ITEM. TO BE FINALISED DURING DETAIL ENGINEERING.
7. TYPICAL XV SYMBOL ON PID TO BE READ AS BELOW.



8. FOR ALL INLINE INSTRUMENTS, UPSTREAM AND DOWNSTREAM DISTANCE TO BE CONSIDERED IN PIPING.
9. ORIFICE TO BE INSTALLED IN HORIZONTAL LINE ONLY.
10. INSTRUMENT ISOLATION VALVES & DRAIN / VENT REQUIREMENT TO BE UPDATED DURING DETAILED DESIGN ALONG WITH THE P&ID.

NOTES:

1. AGITATOR MOTOR SHALL TRIP FROM DCS IN CASE OF MECHANICAL SEAL LEAKAGE.
2. FLAME ARRESTOR WILL BE AN INTEGRAL PART OF PVRV. SAME WILL BE PROCURED ALONG WITH PVRV.
3. FUNNEL ARRANGEMENT & ROUTING TO BE DONE DURING DETAILED ENGINEERING.

HOLDS:

1. PUMP SUCTION AND DISCHARGE NOZZLE SIZE.
2. PVRV SIZE.

No.	DATE	REVISION	DRAWN	CHD	REVD	APPD
0	10-JUL-26	ISSUED FOR DETAIL ENGG.	DVS	NC	DP	KV
B	06-JUN-26	ISSUED FOR DESIGN	DVS	NC	DP	KV
A	29-APR-26	ISSUED FOR APPROVAL	DVS	NC	DP	KV

Rasayan Engineering Solutions

CLIENT: EPIGRAL Limited

PROJECT: BASIC ENGINEERING FOR EPAM PROJECT

TITLE: PIPING & INSTRUMENTATION DIAGRAM BPA+ECH SYSTEM AND PREMIX REACTOR (LER)

SCALE : NTS | DWG.NO.: EPAM01-PR-PID-0101 | SHT: 1 OF 1 | REV. 0